

How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound

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Abstract

Reduced (modal) deformable bodies resting in unilateral contact with rigid bodies are usually resolved by a velocity-level complementarity law that couples rigid and modal unknowns through one shared multiplier. Interactive solvers run that law at small, fixed local iteration budgets. We measure what this costs. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene $\times$ relaxation $\times$ budget sweep, we find sharply formulation-dependent behaviour: the position-based host amplifies modal energy to 1.2×10^5 times the incident rigid kinetic energy in the most starved cells, the augmented-Lagrangian host exceeds that incident-energy benchmark only marginally ($1.7\times$) and only at the starved corner of one scene, and the implicit realization stays below it in every cell (worst 0.53). A budget sweep shows the amplification is a truncation artifact: it decays monotonically toward a converged reference computed in the same code path, and the gap to that reference shrinks by six orders of magnitude. Because the converged regime is unaffordable at interactive budgets, we add a cumulative, source-referenced modal-storage bound, enforced by a reservoir ledger and a projection of the realized modal state, which holds in all 72 measured cells (60 distinct configurations). We also measure what that projection costs contact validity, and find worst-case post-projection penetration of 21.6 mm — 72% of the supporting board’s thickness — with corrective impulses up to $8.7\times$ steady state.

CCS Concepts

• Computing methodologies $\rightarrow$ Physical simulation.

Keywords

contact, modal reduction, passivity, position-based dynamics

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1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with rigid bodies, the established treatment resolves rigid and modal

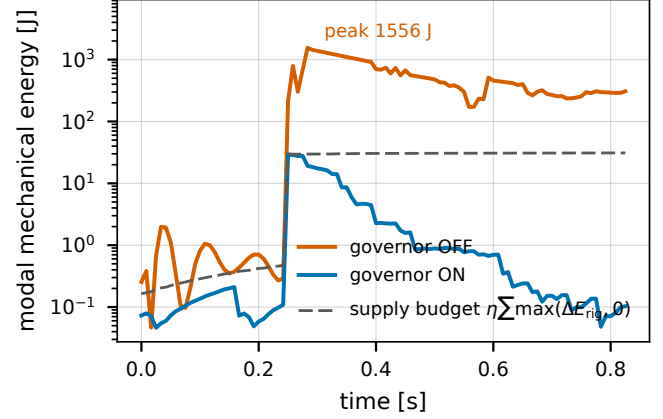


Figure 1: The same impact at the same starved budget (8×2), with and without the cumulative storage bound. The traces are identical until contact. Un-governed, modal energy runs to 1556 J in a scene whose incident rigid kinetic energy is ~ 29 J; governed, it peaks at 29.3 J and stays under the running supply budget that funds it (dashed), which is measured rigid-side contact dissipation, not a tuned constant.

unknowns together through a shared contact multiplier at the velocity level [6]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [1, 2].

We take that coupled row as given. Written at position level it is one linearization step from the velocity-level law: for gap function C , $C^{n+1} \approx C^n + h\mathbf{J}\mathbf{u}^{n+1}$, which divided by h is the velocity-level complementarity condition with a C^n/h stabilization bias, carrying the same Jacobian and the same multiplier as Zheng and James [6]. *The row is not our contribution*, and ours is narrower than theirs: modes couple through normal rows only, and the position-level form takes $e = 0$ restitution.

Interactive solvers do not solve that row to convergence. They run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [4]. *That observation is not ours either*. What is missing is a quantitative answer for this specific case — unilateral contact-to-modal transfer — and a mitigation that fits inside an interactive budget.

Contributions.

- (1) **Measurement.** One contact row, three fixed-budget solver formulations, one identical 24-cell adversarial sweep, one machine (§3.1); plus an iteration-budget sweep against a

converged reference showing the injection is a truncation artifact (§3.2).

- (2) **A cumulative storage bound.** A one-directional, source-referenced bound on energy entering modal storage, enforced by a reservoir ledger and a projection of the realized modal state (§2); it satisfies the prescribed inequality in all 72 measured cells (60 distinct configurations).
- (3) **The cost of enforcement.** Post-projection contact validity — penetration, corrective impulse, multiplier variance, momentum — measured rather than assumed (§3.3).

Relation to concurrent energy-control work. Wei et al. [4] guarantee discrete passivity for any finite inner-iteration budget by construction, using Douglas–Rachford splitting in wave coordinates whose Fejér monotonicity supplies algorithmic dissipation; their interfaces are *bilateral* couplings between subsystems, not unilateral contact, and their guarantee is structural rather than a budgeted transfer cap. You et al. [5] modify implicit Euler to track user-specified *total-energy* targets bidirectionally (dissipation or conservation) on full nonlinear elastodynamics with barrier contact. Ours is neither: a one-directional cap on how much energy contact may deposit into modal storage, funded by measured rigid-side dissipation, enforced by scaling realized modal state. For flexible bodies inside a game engine, Zobel et al. [7] solve ray-traced penalty contact quasi-statically in Unity, without a shared unilateral multiplier; reduced models built from vibration modes and modal derivatives have been used inside a position-based host as a *deformation* surrogate [3], whereas here the modal amplitudes enter as contact DOFs driven by the shared multiplier.

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law.

The invariant. We require that the energy stored in the modal subsystem never exceed what contact has actually dissipated on the rigid side, cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1], \quad (2)$$

where ΔE_{rig}^k is the rigid kinetic energy lost in substep k , gravity work removed so that free fall cannot fund the modes. The transfer efficiency η sets how much of the measured loss may be spent; every experiment in this paper runs $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total stays inside the budget, which is what lets a legitimate ring persist across substeps.

Enforcement. A reservoir accumulates $\eta \max(\Delta E_{\text{rig}}, 0)$ each substep. After the contact solve, if the realized modal energy would

overdraw the reservoir, the realized state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma (\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma \in (0, 1], \quad (3)$$

chosen so that (2) holds with equality at worst. Two consequences matter and are measured below: γ scales energy quadratically, so the projection is inert whenever the solve is already within budget ($\gamma = 1$, bitwise-identical trajectories); and it acts on state *after* the contact solve, so it moves the deformed surface of (1) without re-solving contact — the cost quantified in §3.3.

3 Results

Every solver-behaviour number below was produced on one machine (Apple M4, CPU only, CPython 3.12); the sole exception is the device-resident timing paragraph of §3.5, which is measured on an NVIDIA RTX 4090 and reported separately. All are frozen with their generating commands and commit hashes in the supplemental ledger. Cross-machine mixing is avoided deliberately: chaotic contact stacks diverge between architectures under floating-point reassociation.

3.1 One row, three formulations, the same 24 cells

We drive an impactor into a modally-reduced support in three scenes — a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [1] — and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. The reported ratio is peak modal mechanical energy over peak incident rigid kinetic energy of the impactor: an intensity diagnostic, not the enforced invariant.

Figure 2 shows the outcome. The position-based host injects in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1 : peak modal energy 4.44×10^7 J against an incident 371 J). The augmented-Lagrangian host injects in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is *not* unconditionally benign. The ordering between the two hosts, however, is scene-dependent, and it inverts: taken worst-over-scenes at 4×1 the position-based host is some five orders of magnitude more energetic (1.2×10^5 vs 1.70), but *on the table scene itself* the augmented-Lagrangian host is the worse of the two (1.18 and 1.70 against a position-based worst of 0.37, which does not inject there at all). Neither host dominates the other across scenes; what the sweep establishes is that both are budget-sensitive and the implicit realization is not. That realization stays below the incident-energy threshold in every cell (worst 0.53), and its ledger accounting, which runs as a monitor even with enforcement disabled, confirms the cumulative inequality of (2) in all 24 — the only one of the three hosts for which an un-governed (2) verdict exists.

With enforcement enabled, all 72 cells satisfy (2) — 60 of them distinct configurations, since the implicit host does not consume the relaxation axis and its 12 relaxation pairs are bit-identical by construction (Fig. 2). The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell. At the well-resolved budgets ($16 \times 4, 32 \times 8$) the projection fires zero times on the position-based host at relaxation 0.7: it is inert exactly where the solve is already within budget.

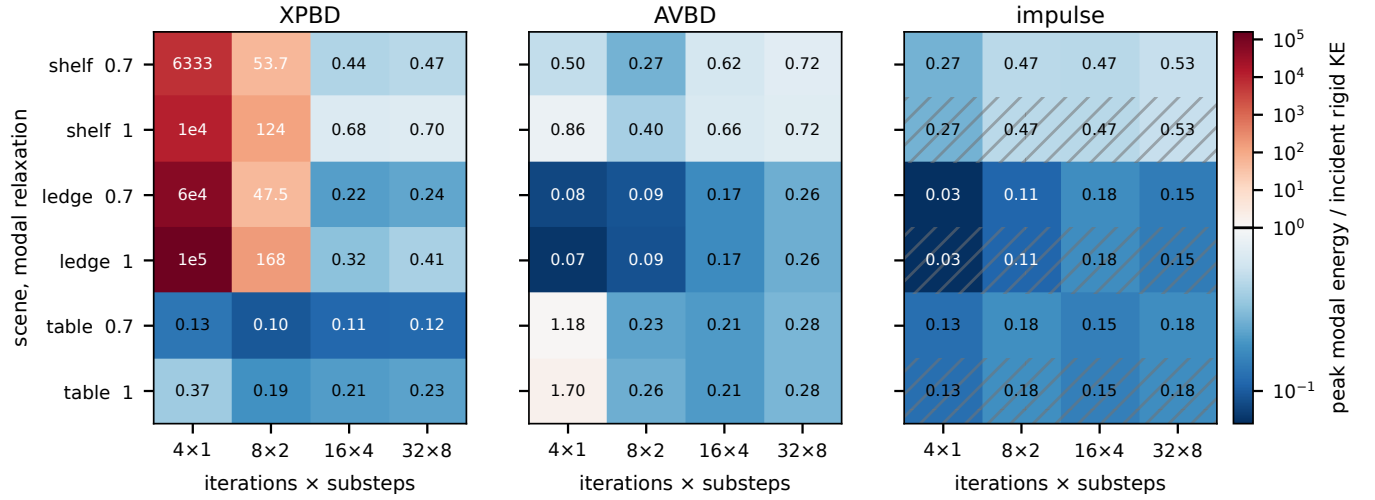


Figure 2: The same contact row in three fixed-budget solver formulations over an identical 24-cell sweep (3 scenes × modal relaxation {0.7, 1.0} × budget), governor OFF. Colour is peak modal energy divided by peak incident rigid kinetic energy, log scale diverging at the incident-energy threshold 1 (white). XPBD injects in 8/24 cells, AVBD in 2/24, the implicit sequential-impulse realization in 0/24. Hatched rows on the right panel mark the relaxation axis, which that formulation does not consume (its implicit modal weight requires no under-relaxation); those rows are bit-identical by construction and are shown for grid symmetry, not as an independent measurement.

3.2 The amplification is a truncation artifact

Figure 3 isolates the local iteration budget. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the converged reference 0.2735; the gap $|\text{XPBD} - \text{reference}|$ shrinks from 2.96×10^4 to 2.7×10^{-2} . The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (spread 4.9×10^{-4}).

This is the central diagnostic result, and it cuts both ways. It confirms that the amplification is created by truncation and cured by convergence — but it equally says that iterating harder, or adopting the implicit formulation, would remove the problem outright. The bound of §2 is therefore not offered as a substitute for convergence. It is offered for the regime where neither escape is available: interactive position-based solvers ship budgets in the neighbourhood of 1×8 to 2×4 , well below the $K \geq 24$ this scene needs, and the position-based formulation is what is deployed. We note explicitly that our implicit backend is a reference implementation whose cost we do not claim to be representative of that formulation class.

3.3 What the projection costs contact

Equation (3) scales $\mathbf{q}$ after the contact solve has finished, which moves the surface in (1) without re-solving contact. Since $\gamma < 1$ shrinks the sag $U_y^T \mathbf{q}$, it lifts the support surface *into* the resting body. Table 1 reports the consequence, instrumented by runtime wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.25 mm, but the worst case reaches 21.6 mm, and the projection increases worst-case penetration by roughly 3.5–12.6× over its pre-scale value. That worst case is large in absolute terms and should be read against the supporting geometry: the shelf board is 30 mm thick and spans

Table 1: Post-projection contact validity, position-based host at starved budgets, per clamp-active substep. Gap violation is measured immediately after the γ -scale; the corrective impulse is the next substep’s normal impulse against the unclamped steady-state median. The 4×1 cells clamp on nearly every substep, so no steady state exists there (—).

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
shelf	4×1	107/108	1.25 / 21.6	—	—
shelf	8×2	162/216	0.30 / 6.46	6.90×	26×
ledge	4×1	98/108	0.07 / 21.2	1.03×	1.07×
ledge	8×2	96/216	0.07 / 4.08	8.69×	58×

0.8 m, so 21.6 mm is 72% of the board thickness and 2.7% of its span. It is a visible interpenetration, not a sub-millimetre artifact, and it occurs in the cell where the projection is most load-bearing (4×1, clamping on 107 of 108 substeps). Contact does recover: the violation is corrected in the following substep rather than accumulating, but that correction costs a normal impulse up to 8.7× the steady-state value, with multiplier variance up to 58× higher on clamp-active substeps. Net rigid linear momentum is unchanged across the projection to exactly zero in every clamp-active substep of every cell — necessarily so, since (3) writes only modal state; we verified it numerically rather than asserting it, and substep-to-substep momentum change on clamp-active substeps stays at or below the unclamped baseline.

3.4 The reduced response against a full-FEM reference

Bounding the energy is only useful if the response being bounded is the right one. On the ledge scene we compare the reduced arm

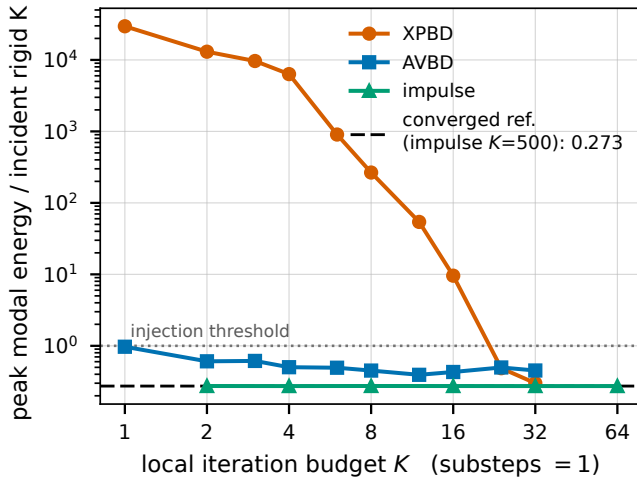


Figure 3: Energy ratio versus local iteration budget K on a deterministic shelf drop, substeps pinned at 1 so K is the only variable, governor OFF. The position-based host is monotone non-increasing in K and decays toward the converged reference, crossing the incident-energy threshold between $K=16$ and $K=24$; the gap to that reference shrinks by six orders of magnitude. The reference is the implicit realization at $K=500$ — the same code path run to convergence, not a different model. That realization is already converged at $K=2$. The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small K , so it must not be read as a convergence trend.

against an unreduced FEM solve of the same mesh, with both arms built on the *same* discrete operator (the reduced basis is eigsh(K, M) of the operator the reference integrates), so discretization bias cancels from every ratio. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as the step refines $h = 1/120 \rightarrow 1/960$ and plateaus near 0.9; the residual $\sim 10\%$ is $k=24$ mode truncation on a thick, high-frequency slab, not a coupling error — the ring frequency already agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees as well: peak deflection sits at the pedestal antinode rather than under the impact, and the falloff along the slab tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term. This is the pre-topple, small-displacement regime; we claim nothing beyond it.

3.5 Cost

The ledger adds 0.8–2.9 ms/step at the evaluated configuration on the CPU host. A separate device-resident path (NVIDIA RTX 4090), which carries the ledger read-only as a monitor rather than enforcing it, reaches 5.0–8.9 ms/step at 16×4 across four scenes — the two above plus a 2.5×1.5 m road slab under a dropped crate and the table scene. The shelf and ledge scenes meet a 120 Hz budget there with margin (1.7 $\times$ and 1.3 $\times$); the road scene sits at the boundary (1.01 $\times$ on the mean, with a 9.3 ms worst step); the table scene does not (0.94 $\times$). All four meet it at 16×2 or below. We make

no unqualified real-time claim, and the governor is not enforced on that path.

4 Limitations

Stable is not accurate. When the projection is load-bearing it damps the modal response toward the budget, and §3.3 shows it opens contact violations of up to 21.6 mm — 72% of the supporting board’s thickness — in the most starved cells. The result is a bounded, non-diverging simulation, not a faithful one; the honest use of the bound is as a safety envelope for budgets that would otherwise diverge.

Convergence is the better cure where affordable. Section 3.2 shows the pathology disappears with iteration count and does not appear at all in the implicit realization. Our timing for that realization is a CPU reference implementation and should not be read as the cost of that formulation class.

Scope of the row. Normal rows only — friction is body-local, so scraping and rolling transfer are outside this coupling — and the position-level form fixes $e = 0$.

The bound governs modal storage only. It constrains energy entering the modal subsystem, not energy created spuriously in rigid degrees of freedom. In the impulse realization we observe box–box rows rectifying a support ring into net rigid motion; the ledger is blind to this by design.

Enforcement is host-side, single-machine, and single-precision-free (CPU float64). A device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work.

Formulation coverage. Three formulations, three scenes, one contact regime. The augmented-Lagrangian result in particular (2/24 cells, worst 1.7 $\times$) shows that “descent-class solvers are passive here” is an empirical statement about this sweep, not a theorem.

5 Conclusion

Transcribing an established rigid–modal contact law into fixed-budget interactive solvers is not energy-safe, and how unsafe it is depends strongly on the solver formulation: we measure 1.2×10^5 , 1.70, and 0.53 as the worst-case ratios of modal energy to incident rigid kinetic energy across position-based, augmented-Lagrangian, and implicit sequential-impulse hosts on the same 24 cells. The amplification is a truncation artifact that convergence removes. Where convergence is unaffordable, a cumulative, source-referenced storage bound enforced by state projection holds in every measured cell — at a contact-accuracy cost we quantify rather than omit.

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